Modeling the XOR Function using MP Neurons and using Perceptrons

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1 XOR using MP Neurons

1. General Model of the MP Neuron

The McCulloch-Pitts (MP) neuron with excitatory and inhibitory inputs is formally defined as:

$$y = \begin{cases} 1 & \text{if } \sum_{i=1}^{n} x_i \geq \theta \quad \text{and} \quad I = 0 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where $x_i \in \{0, 1\}$ are the excitatory inputs, $\theta \in \mathbb{Z}^+$ represents the activation threshold, and $I \in \{0, 1\}$ indicates the presence of an inhibitory input that cancels activation.

2. Logical Decomposition

Since the XOR ($\oplus$) function is not linearly separable by a single neuron, it is expressed in terms of primary logic operations:

$$\text{XOR}(x_1, x_2) = (x_1 \vee x_2) \wedge \neg(x_1 \wedge x_2) \tag{2}$$

3. Network Architecture

We construct a network with three MP neurons organized into two layers:

Hidden Layer

- **Neuron z_1 (OR Function):** Receives x_1, x_2 as excitatory inputs with threshold $\theta_1 = 1$.

$$z_1 = \begin{cases} 1 & \text{if } x_1 + x_2 \geq 1 \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

- **Neuron z_2 (AND Function):** Receives x_1, x_2 as excitatory inputs with threshold $\theta_2 = 2$.

$$z_2 = \begin{cases} 1 & \text{if } x_1 + x_2 \geq 2 \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Output Layer

- **Neuron y (Output with Inhibition):** Receives z_1 as an excitatory input with threshold $\theta_y = 1$, and the output z_2 acts as an inhibitory input $I = z_2$.

$$y = \begin{cases} 1 & \text{if } z_1 \geq 1 \quad \text{and} \quad z_2 = 0 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

4. Truth Table and Evaluation

x_1	x_2	$z_1 = \text{OR}(x_1, x_2)$	$z_2 = \text{AND}(x_1, x_2)$	$y = \text{XOR}(x_1, x_2)$
0	0	0	0	0
1	0	1	0	1
0	1	1	0	1
1	1	1	1	0

Table 1: Evaluation of the three-neuron McCulloch-Pitts circuit.

2 XOR using Perceptrons

2.1 Network Architecture

Let $x_1, x_2 \in \{0, 1\}$ be the binary input signals. The threshold activation function used by each perceptron unit is defined as the step function $f : \mathbb{R} \to \{0, 1\}$:

$$f(z) = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases} \tag{6}$$

where $z = \mathbf{w}^T \mathbf{x} + b = w_1 x_1 + w_2 x_2 + b$.

2.2 Perceptron Weights and Biases

2.2.1 Hidden Layer Unit 1: OR Gate (h_1)

The first hidden neuron acts as an OR gate, activating if at least one input is 1.

- **Weights:** $w_{11} = 1.0, \quad w_{12} = 1.0$
- **Bias:** $b_1 = -0.5$

The net input and output are defined as:

$$z_1 = 1.0 \cdot x_1 + 1.0 \cdot x_2 - 0.5, \quad h_1 = f(z_1) \tag{7}$$

2.2.2 Hidden Layer Unit 2: NAND Gate (h_2)

The second hidden neuron acts as a NAND gate, activating for all input pairs except $(1, 1)$.

- **Weights:** $w_{21} = -1.0, \quad w_{22} = -1.0$
- **Bias:** $b_2 = 1.5$

The net input and output are defined as:

$$z_2 = -1.0 \cdot x_1 - 1.0 \cdot x_2 + 1.5, \quad h_2 = f(z_2) \tag{8}$$

2.2.3 Output Layer Unit: AND Gate ($\hat{y}$)

The output neuron receives h_1 and h_2 as inputs and computes an AND operation, activating only when both hidden units are active.

- **Weights:** $w_{out,1} = 1.0, \quad w_{out,2} = 1.0$
- **Bias:** $b_{out} = -1.5$

The final net input and output prediction are defined as:

$$z_3 = 1.0 \cdot h_1 + 1.0 \cdot h_2 - 1.5, \quad \hat{y} = f(z_3) \tag{9}$$

2.3 Truth Table and Verification

Table 2 demonstrates the evaluation of all four binary input combinations through the two network layers.

Table 2: Step-by-step verification of the XOR Perceptron network.

x_1	x_2	z_1 (OR)	h_1	z_2 (NAND)	h_2	z_3 (AND)	$\hat{y}$ (XOR Output)
0	0	-0.5	0	$+1.5$	1	-0.5	**0**
1	0	$+0.5$	1	$+0.5$	1	$+0.5$	**1**
0	1	$+0.5$	1	$+0.5$	1	$+0.5$	**1**
1	1	$+1.5$	1	-0.5	0	-0.5	**0**